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November 25, 2025

Prof. Paolo Carbone
Editor-in-Chief
Measurement

Dear Prof. Paolo Carbone,

On behalf of my co-authors, I am pleased to submit our manuscript entitled “*Voiceprint Diagnosis Method for Transformer Faults Based on Mel-GADF-PCNN and Improved Physics-Informed Neural Networks*” for consideration for publication in *Measurement*.

Transformer faults diagnosis has traditionally relied on vibration-based methods, whereas studies utilizing acoustic fingerprint-based approaches remain relatively limited. The main challenges in acoustic fingerprint diagnosis are insufficient feature extraction and low recognition accuracy. In this work, we propose a novel multi-feature fusion method for transformer acoustic signals to enhance feature representation and develop a Physics-Informed Neural Network-based model to improve fault recognition performance. Experimental results demonstrate that the proposed approach achieves a recognition accuracy of 98.48%, indicating strong potential for practical applications in transformer operation and maintenance.

Compared with existing data-driven methods, our contributions are threefold:

1. We propose a fusion strategy combining the Mel spectrogram and Gramian Angular Difference Field of transformer voiceprint signals. Gamma correction is applied to enhance the Mel features of the fused image, and a Pulse Coupled Neural Network is employed to improve feature discriminability while suppressing noise interference.
2. We design an improved AlexNet-SE architecture within the PINNs framework to achieve accurate fault diagnosis. Specifically, batch normalization and Squeeze-and-Excitation channel attention modules are integrated after each convolutional layer to refine structural representation and enhance channel feature extraction.
3. Extensive experiments validate the effectiveness of the proposed method, achieving high recognition accuracy and demonstrating robustness compared with conventional approaches.

We confirm that this manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the manuscript and agree with

its submission. There are no conflicts of interest to declare.

We sincerely appreciate your consideration of our manuscript and look forward to your response.

Sincerely,

Zonglong Bai

On behalf of all co-authors

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